

CLASS XI · LECTURE 5

Operating Systems

Two periods · Section C question: define an OS and explain its functions

By the end of this lecture you can

- Define an operating system and state its objectives.
- Explain the main functions in enough detail for a 10-mark answer.
- Distinguish batch processing, multitasking and time sharing.
- Describe booting, and name DOS, Windows and UNIX.

The program that runs the other programs**DEFINITION Operating system**

An operating system is system software that manages hardware and software resources and provides services so that application programs, and the user, can use the computer.

Without an operating system you would have to tell the disk, the memory and the screen how to behave inside every program. The operating system does that work once. Its objectives are convenience for the user, efficient use of the hardware, and a safe place where programs do not casually destroy each other's data.

Names you should be able to place: **DOS** is an older single-user operating system with a command line. **Windows** is a graphical operating system used on most college PCs. **macOS** is Apple's graphical system. **UNIX** is a powerful multi-user system; **Linux** is a UNIX-like system often used on servers. Android, used on phones, belongs with the mobile systems and is mentioned again with Paper I only as an example of an operating system that is not Windows.

Functions — the long answer

Learn eight functions. In Section C, write the definition, then six or seven of these, two sentences each. That is a full answer. A list of bare headings without a verb is half an answer.

Function	What to write
Booting	Loads the operating system into RAM when the computer is switched on, after the BIOS has checked the hardware.
Process management	Starts programs, gives them the CPU in turn, and ends them. Keeps one program from locking the machine.
Memory management	Gives each program a share of RAM and takes it back when the program ends. Stops two programs writing the same bytes.
File management	Creates, names, copies, deletes and finds files. Keeps the folder structure on the disk.
Device or I/O management	Controls keyboard, mouse, printer and disk through drivers, so a program does not need the wiring details.
Secondary storage management	Keeps track of free space and decides where a new file is written.
Protection	Checks user accounts and permissions so that one user cannot read or erase another user's files.
Command interpreter	Reads what the user typed, or the menu the user clicked, and starts the right service. This is the shell or the graphical desktop.
Network management	On a connected machine, handles the network connection and shared resources.

Ways of working

Way of working	Idea	Picture to keep
Batch processing	Jobs are collected and run one after another without the user sitting there	A bank processing a night of cheques
Multitasking	One CPU switches among several programs so they all appear to move	A browser, a player and a document open together
Time sharing	Many users each get a thin slice of CPU time	Several terminals on one host
Multiprocessing	Two or more CPUs share the work inside one machine	A dual-processor server
Real time	A reply must arrive before a deadline	A ticket booking seat, or a hospital monitor

Multitasking is not the same as multiprocessing. Multitasking can happen on **one** CPU by switching. Multiprocessing needs **more than one** CPU. A thread is a smaller strand of work inside a process. You will not be asked to write thread code in Paper I. If a short question appears, say: a process is a running program; a thread is one line of activity inside that process; multitasking switches among programs; multithreading switches among threads inside one program.

The interface, and booting

A **command-line interface** makes you type commands, as DOS and a UNIX shell do. You must know the words, but a skilled user can work quickly. A **graphical user interface** uses windows, icons, menus and a pointer. Windows and macOS are GUI systems. GUI stands for Graphical User Interface, not “general user interface”.

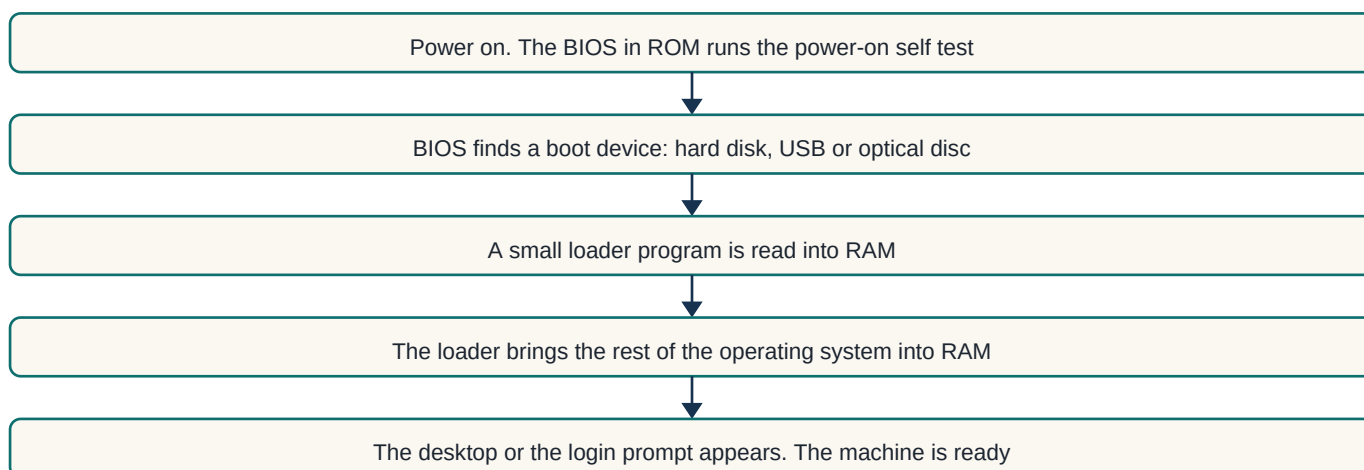


Figure 5.1 Cold boot. A warm boot is a restart while power was already on.

In the Windows practical you should be able to create, rename, copy and delete a folder, recognise a drive letter and a path such as `D:\College\Notes`, search for a file, and see that a file can be read-only or hidden. Those are file-management skills, which is one function of the operating system made visible. The theory paper asks for the function, not for a list of mouse clicks.

Checkpoint

Try the questions before you read the model answers. Section B answers on the real paper should stay within about six or seven lines.

Multiple choice

1. GUI stands for:

- A) General User Integration
- B) Graphical User Interface
- C) Guided Universal Input

- D) General Utility Interface

2. Giving the CPU to several programs in turn on one processor is:

- A) Multiprocessing
- B) Multitasking
- C) Compiling
- D) Modulation

Short answers

S1. What is an operating system? Why is it necessary?

Detailed answer

L1. What is an operating system? Explain its functions.

Model answers

Cover this section while you attempt the checkpoint. The wording is a full-mark style answer, not the only acceptable wording.

MCQ 1. B. Windows, icons, menus and a pointer.

MCQ 2. B. Multiprocessing needs more than one CPU.

S1. An operating system is system software that manages hardware and programs and gives the user a way to command the computer. It is necessary because it manages memory, files and devices, starts and stops programs, and stops every application from having to control the hardware itself. DOS, Windows and UNIX are examples.

L1. An operating system is system software that controls computer resources and provides services to users and application programs. Its functions are as follows. Booting loads the system into RAM at start-up. Process management decides which program uses the CPU. Memory management allocates RAM and reclaims it. File management creates, names, saves and deletes files and folders. Device management controls input and output through drivers. Secondary-storage management tracks free disk space. Protection checks users and permissions. The command interpreter reads commands or clicks and starts the right action. On a networked PC it also manages the connection. Examples are Windows, Linux and UNIX. Without it, application programs could not share one machine safely.